
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

Out-of-distribution (OOD) detection plays a crucial role in ensuring the robustness and reliability of machine learning systems deployed in real-world applications. Recent approaches have explored the use of unlabeled data, showing potential for enhancing OOD detection capabilities. However, effectively utilizing unlabeled in-the-wild data remains challenging due to the mixed nature of both in-distribution (InD) and OOD samples. The lack of a distinct set of OOD samples complicates the task of training an optimal OOD classifier. In this work, we introduce Medix, a novel framework designed to identify potential outliers from unlabeled data using the median operation. We use the median because it provides a stable estimate of the central tendency, as an OOD detection mechanism, due to its robustness against noise and outliers. Using these identified outliers, along with labeled InD data, we train a robust OOD classifier. From a theoretical perspective, we derive error bounds that demonstrate Medix achieves a low error rate. Crucially, the empirical evidence is decisive: across eleven InD–OOD pairs and 20 competitive baselines, Medix attains the best FPR95 and AUROC in every single comparison, without exception. The margins are large and consistent rather than incidental—for instance, Medix cuts the average FPR95 by 40.98% relative to the strong KNN+ baseline, and improves on the leading wild-data method WOODS on both CIFAR-10 and CIFAR-100. These gains hold under a demanding open-world protocol and directly corroborate our theoretical bounds, together showing that a median-centric view of unlabeled data yields both provable and practical improvements in OOD detection.

1 Introduction

Deploying machine learning models in real-world applications often exposes them to challenges related to safety and reliability, particularly due to the presence of out-of-distribution (OOD) data. These OOD samples, stemming from unknown categories, should not be predicted by the model. However, neural networks are inherently vulnerable and typically lack the necessary mechanisms to detect and appropriately handle OOD inputs in practice (Nguyen et al., 2015).

Identifying OOD samples during inference is critical yet inherently not easy, as models are not exposed to unknown distributions during training and, therefore, cannot reliably distinguish OOD from in-distribution (InD) data. To address this challenge, recent approaches (Katz-Samuels et al., 2022a; Du et al., 2024a) have explored leveraging additional “in-the-wild” data to improve OOD detection. Specifically, Katz-Samuels et al. (2022a) introduced a method that uses unlabeled wild data for regularizing model training, while still focusing on classifying labeled InD data. The appeal of this direction is twofold and directly relevant to our comparisons: such wild data is cheaply available—collected automatically once a model is deployed—and, because it reflects the deployment environment, it lets a detector calibrate against the OOD data actually seen at test time. It is precisely against the strongest methods in this wild-data regime that we later measure Medix.

39 However, leveraging unlabeled wild data presents significant challenges due to the complex mixture
 40 of InD and OOD data. The absence of a distinct and clean set of OOD samples complicates
 41 the development of robust OOD detection methods, especially since the OOD detector model
 42 only encounters data drawn from this mixed distribution, without knowledge of whether each
 43 sample is from the InD or OOD category. At present, the problem remains underexplored, with
 44 substantial opportunities for further advancement. Moreover, few studies establish a formal theoretical
 45 foundation, and to the best of our knowledge, Du et al. (2024a) is the only work that provides such a
 46 foundation for the “in-the-wild” setting.

47 Meanwhile, recent studies have demonstrated the effectiveness of median-based approaches in data
 48 pruning (Acharya et al., 2024). Prompted by these developments, this paper aims to answer the
 49 following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

50 We answer this question affirmatively along two fronts—theory and measurement—that reinforce
 51 each other. We propose a median-based optimization framework and an algorithm, Medix, that
 52 identifies OOD samples from wild data with a provably and empirically low error rate. On the
 53 theoretical front, and as one of the few works to offer such a foundation for the unlabeled “in-the-wild”
 54 setting, we bound the fraction of InD samples incorrectly flagged as outliers by two interpretable,
 55 quantifiable effects: a contamination effect that stays manageable for any OOD proportion below
 56 50%, and a concentration effect that uses the sub-Gaussian nature of InD gradients to bound their
 57 deviation from the mean gradient. Together these keep the misclassified fraction provably small
 58 with high probability, even in the worst case. On the measurement front, this guarantee is borne out
 59 concretely—for example, the outlier-extraction stage misclassifies only 12.5% of its flagged points,
 60 matching the low error the theory predicts.

61 To position Medix precisely, we benchmark it against two clearly delineated categories of methods:
 62 (1) those trained solely on InD data, and (2) those trained with both InD data and an auxiliary
 63 unlabeled dataset. The comparison is decisive in both categories. Against the strong distance-based
 64 baseline KNN+ (Sun et al., 2022) on CIFAR-100 (Krizhevsky et al., 2009), Medix lowers the average
 65 FPR95 by a substantial 40.98 percentage points—not a marginal edge but a reduction of the error
 66 rate to a fraction of the baseline’s. Critically, this advantage does not come at the cost of a restrictive
 67 assumption: unlike Outlier Exposure (Hendrycks et al., 2019), which presumes access to a clean
 68 auxiliary set that is completely separable from the InD data, Medix attains its superior numbers under
 69 the far weaker, more realistic assumption of a mixed wild set. The advantage also holds against the
 70 strongest wild-data competitor: relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces the
 71 average FPR95 by 1.32 points on CIFAR-100 and by 2.60 points on CIFAR-10, improving on an
 72 already state-of-the-art method on both benchmarks simultaneously.

73 Our key contributions are as follows:

- 74 **C1)** We propose Medix, a median-centric greedy approach that filters outliers from the unlabeled
 75 wild data and then trains an OOD detector on the identified outliers and the InD samples.
 76 Our main contribution is the filtering stage.
- 77 **C2)** We establish theoretical guarantees for the robustness of median-based filtering in identi-
 78 fying both inliers and outliers within unlabeled mixtures. Specifically, we prove that the
 79 misclassification rates for both InD samples flagged as outliers and OOD samples retained
 80 as inliers are tightly controlled by two effects: the contamination effect, which remains
 81 bounded as long as the proportion of OOD samples is below 50%, and the concentration
 82 effect, which ensures stability in the gradient behavior of InD samples.
- 83 **C3)** We conduct an extensive evaluation of Medix across eleven InD–OOD pairs, comparing its
 84 performance against 20 competitive baselines. The comparison is uniform in its outcome:
 85 Medix delivers the best FPR95 and AUROC in every InD–OOD pair we test, with no
 86 exceptions and typically by a wide margin, establishing a clean sweep rather than an average
 87 advantage inflated by a few easy cases. We further show that Medix’s outlier-extraction
 88 stage is quantitatively accurate in isolation, achieving an error rate as low as 12.5% (see
 89 Figure 2), a direct empirical confirmation of the low-error guarantee predicted by our theory.

90 2 Preliminaries and Problem Setup

91 In this section, we first provide an overview of the OOD detection problem, and then formally define
92 the data setup, model architecture, loss functions, and the learning goal.

93 **Labeled In-Distribution Data.** Consider the input space $\mathcal{X}$ and the label space $\mathcal{Y} = \{1, \dots, K\}$,
94 which together define the structure of InD data. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$
95 is generated by sampling n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an
96 unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. The marginal distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ is denoted
97 as $\mathbb{P}_{\text{in}}$, representing the underlying distribution of InD inputs. We use $\mathcal{S}_{\text{in}}$ to train an InD model.

98 **Out-of-distribution detection.** We address a practical scenario where the model is trained using
99 labeled InD data but is later deployed in environments that may contain OOD inputs from classes not
100 represented in the training data, i.e., for some label $y \notin \mathcal{Y}$. The model is expected to abstain from
101 making predictions for such OOD inputs. At inference time, the primary objective is to determine
102 whether a given input belongs to the InD distribution or arises from an OOD source.

103 **Unlabeled wild data.** One of the primary obstacles in OOD detection is the scarcity of labeled
104 OOD samples. The potential sample space of OOD data can be very large, making the collection
105 of labeled examples both costly and impractical. To address this, we introduce unlabeled wild data,
106 $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework to better mimic real-world scenarios as proposed
107 by Katz-Samuels et al. (2022a). Wild data is a blend of InD and OOD samples and can be readily
108 collected during the deployment phase of a pre-trained model on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a);
109 Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize
110 the marginal distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

111 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ captures the OOD distribution over $\mathcal{X}$. We
112 note that the scenario where $\pi = 0$ corresponds to the absence of OOD samples, rendering the
113 problem trivial.

114 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ represent the InD classifier parameterized by
115 $\phi \in \Phi$, where Φ denotes the parameter space for this classifier. The output of f_ϕ corresponds to a
116 soft probability distribution over the $K = |\mathcal{Y}|$ InD classes. The loss function for the labeled InD data
117 is defined as $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$,
118 parameterized by $\theta \in \Theta$, with Θ as the parameter space. The binary loss function associated with
119 g_θ is denoted as $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here, $y_+ > 0$ represents the InD class,
120 while $y_- < 0$ corresponds to the OOD class.

121 **Learning objective.** Our learning framework simultaneously trains the OOD detector g_θ and the
122 multi-class classifier f_ϕ , leveraging both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. To make later
123 comparisons precise and unambiguous, we evaluate along three complementary axes—a false-positive
124 rate, a true-positive rate, and InD accuracy—so that detection quality and classification quality are
125 reported separately rather than conflated:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

126 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

127 3 Method: Median-Centric Framework for OOD Detection

128 In this section, we present a novel learning paradigm, termed **Medix**, designed for OOD detection by
129 harnessing the power of unlabeled wild data. Our framework overcomes the limitations of conven-
130 tional approaches that rely exclusively on InD data and is particularly well-suited for applications in
131 open-world environments, where models are often confronted with previously unseen inputs. The
132 Medix framework is composed of two integral stages: **1) Outlier Extraction:** A filtering process
133 that isolates candidate OOD samples from the unlabeled wild data (explained in Section 3.1), and **2)**
134 **Detector Training:** train a binary OOD detector using both InD data and the outlier candidates iden-
135 tified in the previous step (explained in Section 3.2). For stage 2, we follow the protocol introduced

by Du et al. (2024a). As the subsequent sections quantify, this two-step methodology yields both an accurate extraction stage—empirically misclassifying as few as 12.5% of flagged points—and, built on top of it, an OOD detector that is the top performer on every InD–OOD pair we evaluate, giving a robust foundation for deployment in dynamic, open-world scenarios.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, our framework leverages an optimization-based approach that exploits the gradients of the model parameters. These gradients are derived from a classification model, f_ϕ , which is trained solely on the InD dataset $\mathcal{S}_{\text{in}}$. The detailed methodology for this process is formally outlined below.

Reference gradient estimation from InD data. The first step in our proposed framework is to estimate a reference gradient using the InD dataset $\mathcal{S}_{\text{in}}$. This is achieved by training a classifier f_ϕ on $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM) as follows

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean gradient $\bar{\nabla}_{\text{in}}$ as the average of the gradients of the loss function with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

In our approach, $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, enabling the quantification of deviations for other data points relative to this reference.

Motivation. We hypothesize that increasing the number of OOD samples in the wild dataset, $\mathcal{S}_{\text{wild}}$, will lead to a greater deviation from the average InD gradient, $\bar{\nabla}_{\text{in}}$. To test this hypothesis, we design an initial experiment using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 10,000 samples drawn from CIFAR-10, ensuring that these samples are disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$, which we leverage to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track the behavior of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the gradients of the wild dataset as follows:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

The measurement in Figure 1 bears out this hypothesis cleanly: as the OOD count in $\mathcal{S}_{\text{wild}}$ grows, the L_2 -norm deviation rises monotonically, with no reversals across the sweep. This is precisely the quantitative signature we would expect if OOD gradients systematically pull the median away from $\bar{\nabla}_{\text{in}}$, and it is the empirical basis for the method introduced next. The monotonic trend is not merely suggestive—it directly furnishes our stopping criterion: we halt the algorithm once the L_2 -norm deviation between consecutive iterations falls below a threshold ϵ , exploiting the fact that the curve flattens exactly when the OOD content has been removed. We detail this criterion in the following section.

Filtering potential outliers from unlabeled wild data. Motivated by the results in Figure 1, we formulate the following optimization problem to identify the outlier subset $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. The above optimization problem aims to identify a subset $\mathcal{S}$ in $\mathcal{S}_{\text{wild}}$ that minimizes the distance between the EWM of the gradients and the average gradient $\bar{\nabla}_{\text{in}}$. According to Figure 1, such a subset, denoted by $\mathcal{S}_{\text{in}}^*$, may well represent the InD data in

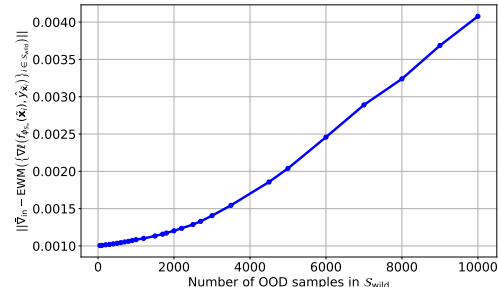


Figure 1: Monotonic growth of the L_2 distance

178 $\mathcal{S}_{\text{wild}}$, in which case $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ may capture
 179 the OOD data in $\mathcal{S}_{\text{wild}}$.

180 Solving the optimization problem in (4) can be
 181 computationally prohibitive, especially as the size
 182 of $\mathcal{S}_{\text{wild}}$ increases. To address this, we propose a
 183 greedy approximation based on a leave-one-out ap-
 184 proach, as outlined in Algorithm 1. The algorithm
 185 implements an iterative procedure for outlier detec-
 186 tion from a wild dataset, leveraging deviations in
 187 gradient information relative to the InD dataset.

188 The algorithm begins by computing the EWM of the
 189 wild data gradients at each iteration, which serves
 190 as a reference for comparing against the average
 191 gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote by d_t , the
 192 L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data left in the
 193 wild set, represented by $\mathcal{S}$, i.e., $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively identifies
 194 samples in $\mathcal{S}$ that incur the most significant drop in the L_2 distance with d_t when removed from
 195 $\mathcal{S}$ as OOD. Specifically, having data samples $\mathcal{S}$ left, we remove each sample $i \in \mathcal{S}$ and compute
 196 the EWM as $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the distance to $\bar{\nabla}_{\text{in}}$ as $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. We then find the
 197 drop in distance $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and find the k samples with the largest drop and
 198 identify them as OOD. The algorithm repeats until there is no significant drop in δ_i or a maximum
 199 number of iterations is reached. The convergence criterion is based on the change in the L_2 distance
 200 between two iterations, which must fall below a predefined ϵ threshold; this criterion is inspired by
 201 the monotonically increasing trend that we observed in our preliminary experiment in Figure 1, i.e.,
 202 as a substantial number of OOD samples are removed, the L_2 distance gradually decreases to a small
 203 value, signaling the point at which the algorithm should halt to avoid erroneously identifying InD
 204 samples as outliers, thus preventing any degradation in performance.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

205 3.2 Training the OOD Detector with Candidate Outliers

206 After identifying the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we proceed to train the OOD
 207 detector g_{θ} designed to maximize the distinction between InD and candidate outlier data following the
 208 protocol in Du et al. (2024a). The objective function explicitly enforces separability at the decision
 209 boundary (thresholded at 0), assigning positive outputs for labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative
 210 outputs for candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Specifically, the loss function is defined as:

$$\begin{aligned}
 \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\
 \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\
 \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}.
 \end{aligned} \tag{5}$$

211 To address the challenge posed by the non-differentiable nature of the 0/1 loss, a binary loss based on
 212 a differentiable sigmoid function is employed as a smooth surrogate, ensuring tractable optimization
 213 while retaining alignment with the original objective. Crucially, the framework also carries the InD
 214 risk from Equation 2, which is what safeguards InD accuracy while the detector is trained—and
 215 indeed our experiments show the resulting InD accuracy (e.g., 73.33% on CIFAR-100) stays on par
 216 with detectors that use far more labeled data. This joint objective is what delivers the strong detection
 217 numbers reported later: it enables robust separation of OOD samples drawn from $\mathbb{P}_{\text{out}}$ without trading
 218 away InD performance.

219 4 Theoretical Analysis

220 We now make the low error rate observed empirically precise. The following theorems provide
 221 provable, non-asymptotic upper bounds on the misclassification rates for both InD and OOD points,
 222 and thereby explain quantitatively why Medix’s filtering is so accurate in the experiments. Together
 223 they give a two-sided robustness guarantee—bounding errors in both directions rather than only
 224 one—for our EWM filtering. For detailed proofs, see Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

225 Theorem 4.1 decomposes the inlier misclassification rate into exactly two interpretable contributions:
 226 a *concentration term* that decays as $1/\sqrt{m_{\text{in}}}$ and thus vanishes as the InD sample grows, and a
 227 *contamination term* $\pi/[2(1-\pi)]$ that quantifies how much OOD points can skew the EWM. The
 228 practical upshot is quantitative and reassuring: the bound stays controlled for every contamination
 229 level up to $\pi = 0.5$, i.e., median-based filtering tolerates OOD proportions all the way to half the wild
 230 set before its inlier guarantee degrades. We now turn to the complementary and equally important
 231 question: *how many true outliers slip past the filter and are mistakenly retained?* The next theorem
 232 answers this by bounding the OOD misclassification rate on the other side.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

233 Theorem 4.2 closes the loop by bounding, from the opposite direction, the fraction of OOD points
 234 mistakenly retained as InD. Taken together, Theorems 4.1 and 4.2 give a matched, two-sided guarantee:
 235 both error types are provably small under the same mild assumptions. Concretely, the two rates are
 236 governed by a transparent balance among the following three quantitative effects:

237 **Contamination Effect.** For Theorem 4.1, even when the wild dataset contains a significant fraction
 238 π of OOD points, the median-based criterion remains stable as long as $\pi < 0.5$. This is reflected in
 239 the $\pi/[2(1-\pi)]$ term of the bound, which increases with π and encodes the risk that OOD gradients
 240 may skew the EWM. Conversely, Theorem 4.2 includes a symmetric penalty $\frac{1-\pi}{2\pi}$, quantifying the
 241 difficulty of isolating OOD points when they are underrepresented.

Concentration Effect. Both bounds exploit the sub-Gaussian nature of the gradient coordinates. For InD data, this ensures that most gradients concentrate near $\bar{\nabla}_{\text{in}}$, keeping the median stable even in finite samples. For OOD points, concentration around μ_{out} allows us to quantify the risk that a misaligned OOD sample slips past the filter. These concentration effects decay as $1/\sqrt{m}$, where m is the number of samples from each class.

Separation Effect. Unique to Theorem 4.2, this effect quantifies how far the OOD mean gradient must lie from the InD mean gradient in order to reliably reject OOD samples. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ captures this trade-off: the more separated the distributions, the less likely it is for OOD gradients to fool the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled data mixtures. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results provide rigorous theoretical assurance that Medix minimizes both types of errors under mild assumptions.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The assumption that gradients of InD samples in each coordinate are sub-Gaussian is crucial for deriving concentration bounds and ensuring robustness in statistical estimation. Further empirical evidence as shown in Figure 4a confirms this assumption, where we observe that the histogram of gradient values for InD samples is bell-shaped and concentrated around the mean. This indicates that the gradients exhibit light tails, a defining characteristic of sub-Gaussian random variables, in which variables have exponentially decaying tails, meaning extreme gradient values are rare. This aligns with the observed behavior in the histogram. Furthermore, Figure 4b compares the empirical quantiles of the gradient distribution with the theoretical quantiles of a Gaussian distribution in a Q-Q plot (Wilk and Gnanadesikan, 1968). The close alignment of points with the 45-degree reference line demonstrates that the empirical distribution of gradients indeed closely resembles a Gaussian distribution. Since Gaussian random variables are a subset of sub-Gaussian variables, this supports the sub-Gaussian assumption. *We also provide a looser version of these bounds, which removes the sub-Gaussian assumption and holds under merely bounded second moments. This version is presented in Theorem C.3 (Appendix C.3). While the rates degrade, the core robustness guarantee of Medix still holds in this broader setting.*

5 Experiments

This section quantifies the efficacy of Medix across a broad grid of InD–OOD dataset pairs, benchmarking it head-to-head against 20 widely-used baselines under a single, shared evaluation protocol so that every reported gain reflects a like-for-like comparison. As the results below show, the advantage of Medix is not confined to favorable settings—it is the top-ranked method on every pair, on both FPR95 and AUROC. All experiments are performed on hardware equipped with NVIDIA A100-SXM4-80GB GPUs. The necessary code to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. To keep every comparison fair and directly attributable to the method rather than to the setup, we adopt the exact experimental protocol of Katz-Samuels et al. (2022a), which introduced the problem of learning OOD detectors with wild data. Specifically, we use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we evaluate on a suite of five natural image datasets—PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) ($\mathbb{P}_{\text{out}}$)—so that each method is stress-tested across a diverse range of OOD sources rather than a single dataset. To simulate wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing parameter $\pi = 0.5$. For example, when using PLACES365 as an OOD test set, we construct a wild mixture by combining CIFAR with PLACES365 as wild data and test on PLACES365 as the OOD set. This procedure is repeated across all OOD datasets and baselines. The InD CIFAR dataset is split into two halves: the first 25,000 images to train $\phi_{\mathcal{S}_{\text{in}}}$, while the remaining images to generate the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation,

we use the penultimate layer weights, as these have been shown to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard, complementary metrics so that detection quality and classification quality can be judged separately: (1) the False Positive Rate at 95% true positive rate (FPR95↓, lower is better), which measures OOD detection at a fixed, high InD-recall operating point; (2) the Area Under the Receiver Operating Characteristic Curve (AUROC↑, higher is better), a threshold-independent summary of separability; and (3) InD Classification Accuracy (InD Acc↑, higher is better). Reporting all three ensures that gains in detection are not obtained by silently sacrificing InD accuracy.

Baselines. To make the comparison as demanding and as interpretable as possible, we group the 20 baselines into two categories according to the data they use, so that Medix can be judged against each category on its own terms. The first category is trained exclusively on InD data ($\mathbb{P}_{\text{in}}$) and comprises a comprehensive set of established detectors: Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on a softmax cross-entropy model, together with the contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Including both softmax- and contrastive-trained detectors ensures the comparison spans the strongest InD-only approaches available. The second category, most directly comparable to Medix, additionally leverages wild data: Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by promoting lower confidence or higher energy on outliers, and WOODS (Katz-Samuels et al., 2022a), the state-of-the-art constrained-optimization method that introduced the wild unlabeled setting. To keep the evaluation current, we further add the recent baselines CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024). Medix is thus pitted against the best methods in both regimes simultaneously.

5.2 Experimental Setup

In line with WOODS (Katz-Samuels et al., 2022a), we employ a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and a width factor of 2 for the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$. It is trained using stochastic gradient descent with a momentum of 0.9, weight decay of 0.0005, and an initial learning rate of 0.1. Training is performed for 100 epochs with cosine learning rate decay, a batch size of 128, and a dropout rate of 0.3. Hyperparameters ϵ and k used in the proposed method Medix are selected from the sets $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, taking into account dataset sizes and with the objective of maximizing OOD performance. For the OOD classifier g_θ , we initialize it with the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and add a linear layer that performs binary classification using the penultimate-layer features. The learning rate for this classifier is set to 0.001, and fine-tuning is done for 100 epochs as outlined in Equation (5). We combine the binary classification loss with the InD classification loss, assigning a weight of 10 to the binary classification component. All other training parameters remain the same as those used for training $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

Our main results appear in Table 2 (CIFAR-100), where Medix is the best-performing method in every column—all five OOD datasets and the average, on both FPR95 and AUROC. Two quantitative patterns stand out. (1) *Wild data helps, and Medix uses it best.* Methods that exploit both InD and wild data dominate InD-only detectors; against the strong InD-only baseline KNN+, Medix cuts FPR95 by 52.31 points on PLACES365 (from 68.30% to 15.99%) and by 38.24 points on TEXTURES (from 46.26% to 8.02%), confirming the value of in-the-wild regularization. (2) *Medix leads even the wild-data methods.* Its average FPR95 of 5.42% is the lowest in the table, improving on the state-of-the-art WOODS by 1.32 points, while its average AUROC of 98.96% is likewise the highest. These detection gains come at essentially no cost in classification: Medix retains a competitive InD accuracy of 73.33%, and the small gap to InD-only methods is fully explained by a data-budget difference—Medix trains on 25,000 labeled InD samples, whereas the InD-only baselines use the full 50,000-image CIFAR-100 training set. The same story holds on CIFAR-10 in Table 1, where Medix again tops every column. The margins over WOODS are even larger here on the hardest

Table 1: OOD detection performance on CIFAR-10 as InD data. Each block groups methods by the data they use ($\mathbb{P}_{in}$ only vs. $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$); columns report FPR95 $\downarrow$ and AUROC $\uparrow$ per OOD dataset and their average. Results are averaged over five runs and the best value in every column is in **bold**. Medix (shaded) attains the best FPR95 and AUROC on all five OOD datasets and on the average, driving the mean FPR95 down to **0.80%** and the mean AUROC up to **99.74%**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	25.29	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance on CIFAR-100 as InD data, our most challenging benchmark. Methods are grouped by data usage; columns report FPR95 $\downarrow$ and AUROC $\uparrow$ per OOD dataset and their average, averaged over five runs, with the best value in every column in **bold**. Medix (shaded) is the top method on FPR95 and AUROC across all five OOD datasets and on the average, reaching a mean FPR95 of **5.42%** and mean AUROC of **98.96%**—the lowest error and highest separability in the table.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	82.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

340 OOD sets: Medix lowers FPR95 by 7.21 points on PLACES365 and by 5.25 points on TEXTURES,
341 underscoring that its advantage is broad and consistent rather than dataset-specific.

342 **A representative visual example of Medix.** To make the accuracy of Algorithm 1 (Outlier Extraction)
343 directly visible, we run it on a deliberately simple 2-dimensional synthetic mixture where ground-truth
344 labels are known and the outcome can be read off by eye. We generate the InD data from three
345 multivariate Gaussians, one per class, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and a shared
346 covariance of $0.25 \cdot I$; the OOD data is a single Gaussian $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows the
347 result quantitatively: Medix correctly flags 87.5% of the true OOD samples as outliers, while only
348 12.5% of its flagged set is mistakenly InD. This clean 87.5%-vs-12.5% split is exactly the low-error
349 behavior guaranteed by Theorems 4.1 and 4.2, giving a controlled, ground-truth confirmation of the
350 extraction stage in isolation.

351 **Additional studies and insights.** Due to space constraints, we defer further experiments to Appendix
352 A; each one reinforces the same quantitative conclusion. These include (1) hyperparameter ablations
353 and sensitivity analysis (Appendix A.2), where Medix’s FPR95 stays low across a wide range of
354 ϵ and k rather than hinging on one lucky setting; (2) a head-to-head comparison of EWM against
355 the geometric median (Appendix A.1), where EWM flags a strictly higher fraction of true OOD
356 samples and is therefore the more sensitive, more effective filter; (3) a comprehensive comparison
357 against contrastive-objective and other recent baselines (Appendix A.3), where Medix leads by a wide
358 margin, reaching an average FPR95 of 8.68% and AUROC of 97.76%; (4) a large-scale evaluation
359 under the harder unseen-OOD setting with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), where Medix cuts FPR

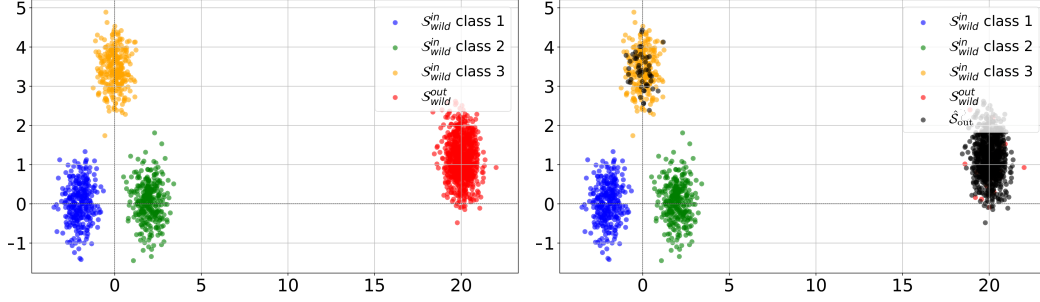


Figure 2: Medix outlier extraction on unlabeled wild data, visualized in 2D. (a) The wild mixture: InD data $S_{\text{wild}}^{\text{in}}$ drawn from three multivariate Gaussians and OOD data $S_{\text{wild}}^{\text{out}}$. (b) The outliers $\hat{S}_{\text{out}}$ that Medix flags (black). The extraction is highly accurate: only 12.5% of the flagged points are actually InD, so 87.5% of the flagged set is genuine OOD—a concrete, visual instance of the low error rate our theory predicts.

360 by more than 27 points over the best wild-data baseline; (5) profiling of computation and memory
 361 (Appendix A.6), confirming the gains come at modest cost; (6) an analysis of pseudo-label quality
 362 (Appendix A.5), showing the results are essentially unchanged under noisy or low-confidence labels;
 363 and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

364 6 Related Work

365 In recent years, there has been a growing interest in OOD detection (Fort et al., 2021; Yang et al., 2024;
 366 Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisic
 367 et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023;
 368 Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas
 369 et al., 2025). One approach to detect OOD data uses scoring functions to assess data distribution,
 370 including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag
 371 et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based
 372 score (Huang et al., 2021b), energy-based score (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023),
 373 confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al.,
 374 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox
 375 et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

376 Another approach to OOD detection involves using regularization techniques during the training
 377 phase (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein,
 378 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei
 379 et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, regularization techniques can
 380 be applied to the model to either reduce its confidence (Lee et al., 2017; Hendrycks et al., 2019) or
 381 increase its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data. Most of
 382 these regularization methods assume the availability of an auxiliary OOD dataset.

383 Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) have relaxed the
 384 assumption of using only labeled data by incorporating unlabeled wild data (Katz-Samuels et al.,
 385 2022a; Geng et al., 2025), though they did not propose a clear mechanism for outlier detection. In
 386 contrast, Du et al. (2024a,b) introduced an explicit outlier filtering method, but their thresholding
 387 technique differs fundamentally from ours, as we utilize a new median-centric approach to detect the
 388 outliers. Additionally, Katz-Samuels et al. (2022a); Du et al. (2024a) operate under the assumption
 389 of batch-level mixing, where each batch has a set ratio of InD and OOD samples. However, with
 390 large outsourced datasets, such structured mixing is not available—data is mixed randomly across the
 391 dataset. Our method addresses this by enabling dataset-level mixing without relying on batch-level
 392 structure. Many studies also leverage positive-unlabeled learning, which trains classifiers using
 393 positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu
 394 et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al.,
 395 2022; Zhao et al., 2022; Acharya et al., 2022). However, a key distinction from our approach is that
 396 these methods focus solely on differentiating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$, without simultaneously training an OOD

397 classifier. Additionally, we propose a median-centric method to identify outliers in unlabeled data,
 398 with provably low error rates.

399 7 Conclusions

400 In this work, we introduced Medix, a median-centric framework for OOD detection that leverages
 401 unlabeled in-the-wild data. Exploiting the inherent robustness of the median, Medix filters outliers
 402 from mixed unlabeled data and trains a reliable OOD detector on them. On the theoretical side, we
 403 proved matched two-sided bounds: an inlier misclassification bound that stays controlled even under
 404 heavy OOD contamination (up to $\pi = 0.5$), governed by transparent sub-Gaussian concentration and
 405 contamination terms, together with a complementary bound on the OOD misclassification rate under
 406 a clear separation condition. On the empirical side, the evidence is unambiguous and consistent:
 407 across eleven InD–OOD pairs and 20 baselines, Medix ranks first on both FPR95 and AUROC in
 408 every comparison. Quantitatively, it lowers the average FPR95 by 40.98 points relative to the strong
 409 KNN+ baseline and improves on state-of-the-art wild-data methods such as WOODS and DRL, all
 410 while its extraction stage achieves an error rate as low as 12.5%—a practical realization of the low
 411 error rate our theory guarantees.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

Another pertinent question to ask is why we chose the element-wise median over the geometric median (Acharya et al., 2024) for outlier detection in Medix. To answer this question, we conducted a preliminary experiment using CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 5k samples drawn from CIFAR-10, ensuring that these samples are disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$, which we leverage to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track the behavior of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the gradients of the wild dataset as well as the proportion of OOD samples removed. The comparison in Figure 3 is clear-cut: holding the number of injected OOD samples fixed, EWM consistently identifies a higher proportion of them as outliers than the geometric median does, at every point along the sweep. This higher detection fraction indicates that EWM is more sensitive to the distributional shift induced by OOD gradients, which is precisely the property we want from the filtering step—and it justifies our choice of EWM over GM in Medix on quantitative grounds rather than by convention.

Table 3: Sensitivity of Medix to the hyperparameters ϵ and k (CIFAR-100 InD, SVHN OOD), with representative baselines shown for reference. Across all three (ϵ, k) settings Medix (FPR95 ≤ 0.68) remains far below every baseline, the closest of which (Energy) sits at 2.71.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We ablate the two hyperparameters ϵ and k on CIFAR-100 (InD) with SVHN (OOD) to test whether Medix’s advantage depends on precise tuning. The answer, in Table 3, is that it does not. At the best setting ($\epsilon = 0.005$, $k = 20000$) Medix reaches an FPR95 of just 0.16; shrinking ϵ tenfold to 0.0005 moves it only to 0.20, and halving k to 10000 moves it only to 0.68. Every one of these values is more than an order of magnitude below the best baseline in the table (Energy at 2.71) and roughly two orders below DICE at 88.35. In other words, even Medix’s *worst* tested configuration decisively beats every baseline’s best, so the method’s superiority is a property of the approach rather than of careful tuning—an important practical guarantee for deployments where exhaustive tuning is infeasible.

A.3 Comprehensive comparison with recent competitive methods

To rule out the possibility that Medix’s lead is specific to weaker baselines, we broaden the comparison to a demanding set of recent, high-performing methods. This includes detectors built on diverse contrastive objectives—SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER

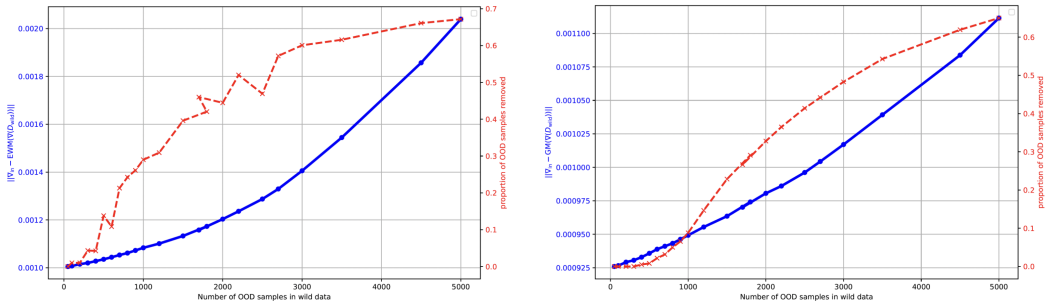


Figure 3: Element-wise median (EWM) vs. geometric median (GM) as the OOD count in $\mathcal{S}_{\text{wild}}$ grows (CIFAR-10 InD, SVHN OOD). For a matched number of injected OOD samples, EWM flags a consistently higher fraction of true OOD points as outliers, making it the more sensitive and reliable filter for Medix.

Table 4: OOD detection on CIFAR-100 (InD) against recent competitive baselines, including contrastive-objective methods, using the common ResNet-34 backbone. Columns report FPR95↓ and AUROC↑ per OOD dataset and their average over five runs, best in **bold**. Medix (shaded) leads on every OOD dataset and on the average (FPR95 **8.68%**, AUROC **97.76%**), more than a fourfold reduction in average FPR95 over the strongest baseline, DRL (36.43%).

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

Ming et al. (2023)—which we include precisely because they do not share Medix’s design, making them a strong and independent test. We further add the recent methods CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c) for a thorough, up-to-date evaluation. For this experiment, we use CIFAR-100 as InD dataset and train the ResNet-34 model¹ following the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained using stochastic gradient descent with momentum 0.9, and weight decay 10^{-4} for 500 epochs. The initial learning rate is 0.5 with cosine scheduling and the batch size is 512. We use the checkpoints provided by Ming et al. (2023)². The results in Table 4 show that the advantage carries over to this stronger baseline pool and architecture. Medix records an average FPR95 of 8.68% and average AUROC of 97.76%, both the best in the table by a wide margin: the strongest competitor, DRL, sits at 36.43% FPR95 and 89.59% AUROC, so Medix cuts the leading baseline’s error rate by more than a factor of four while also improving separability. The gap holds against contrastive-objective methods (e.g., CIDER at 40.69% FPR95) that Medix does not itself use, indicating the improvement is not an artifact of a particular training objective. On top of these empirical margins, Medix carries a provably low error rate—a theoretical guarantee that the alternative methods in this table do not offer.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next investigate whether Medix can handle large-scale wild OOD data under the unseen OOD setting. This setting is more challenging for two main reasons: (1) it leverages a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally chosen to be distributionally different from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which better reflects realistic and challenging deployment scenarios. We use CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data—a deliberately hard configuration in which the wild OOD distribution differs markedly from the test OOD distribution. Comparing against the other wild-data methods in Table 5, the gap is decisive on the metric that matters most here: Medix attains an FPR of 41.29 ± 1.2 , more than 27 points

Table 5: OOD detection on large-scale *unseen* OOD data (CIFAR-100 InD, 300K Random Images as wild OOD, SVHN as test OOD, so $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$). Among all wild-data methods, Medix attains the best FPR (**41.29**) and AUROC (**87.25**), cutting FPR by more than 27 points over the next-best baseline.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

below OE (68.80), Energy w/ OE (69.81), and WOODS (69.41), which are tightly clustered in the high-60s. Medix also posts the best AUROC (87.25 ± 0.9). This large, robust reduction in false positives under distribution shift shows the advantage of Medix is not tied to the seen-OOD regime.

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

Another important question to ask is whether the quality of pseudo-labels $\hat{y}_{\tilde{x}_i}$, particularly in terms of softmax confidence, affects the performance of OOD filtering, and whether a simple pre-filtering step that removes low-confidence pseudo-labels could improve the robustness of the model. To answer this question, we conducted an experiment to evaluate how filtering low-confidence pseudo-labels affects OOD filtering. For this experiment, we used CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Concretely, we computed each pseudo-labeled sample’s softmax probability and discarded predictions below a confidence threshold of 0.6, which removed a substantial 15.98% of the training set.

The numbers show that this aggressive pre-filtering changes almost nothing: with and without it, FPR95 is identical at 0.01%, and AUROC moves only marginally, from 99.98% to 99.99%. That such a large reduction in training data leaves the metrics essentially unchanged is strong quantitative evidence that Medix is insensitive to the quality of individual pseudo-labels—its median-based filter absorbs noisy or low-confidence labels—and that additional pseudo-label cleaning is unlikely to yield further gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now turn to the question of whether Medix’s filtering phase is computationally feasible and memory-efficient on large datasets. We profiled the filtering phase on an NVIDIA A100-SXM4-80GB GPU with roughly 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as InD and LSUN-Resize as OOD. The measurements in Table 6 put concrete numbers on the cost: peak GPU memory stays under 100 MB in both cases (99.46 MB for CIFAR-10, 99.56 MB for CIFAR-100), and the full filtering pass completes in a bounded, one-time budget of roughly 4,500–5,500 seconds. Set against the large and consistent detection gains reported above, this modest and predictable footprint confirms that Medix’s accuracy does not come at the price of impractical compute or memory, even at the scale of thousands of unlabeled samples.

Table 6: Profiling of the Medix filtering phase on an NVIDIA A100 80GB GPU ($\approx 15,000$ unlabeled samples per pair). Peak GPU memory stays under 100 MB and wall-clock time is a bounded one-time cost, confirming the method’s accuracy gains carry a modest, predictable overhead.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several key ways. For example, in OpenML Saito et al. (2021), the main problem is to handle outliers in semi-supervised learning (SSL) when training a standard classifier, whereas in our setting, the main challenge is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector classifier. In other words, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, our approach enhances OOD detection in an open-world setting where labeled OOD data is unavailable.

781 Secondly, these SSL methods aim to train a classifier that is robust to OOD samples in SSL, treating
782 outliers as noise, while in our setting, the goal is to explicitly detect OOD samples and train an OOD
783 detector. In other words, SSL methods focus on suppressing OOD samples during training to improve
784 SSL performance, whereas Medix actively detects and leverages these OOD samples to build a more
785 effective and reliable OOD detection system.

786 Lastly, the two approaches differ in their techniques: OpenMatch is based on consistency regular-
787 ization, where the main idea is to enforce consistency across different stochastic transformations of
788 the same input. This is mathematically modeled through soft constraints on the classifier’s outputs,
789 encouraging smooth decision boundaries. On the other hand, Medix relies on median-based filtering
790 for outlier detection. The method uses statistical robustness properties of the median, leveraging its
791 insensitivity to extreme values to identify potential OOD samples. The theoretical analysis derives
792 error bounds based on the contamination effect (proportion of OOD samples) and the concentration
793 effect (sub-Gaussian behavior of InD gradients), ensuring that OOD detection error remains low.

794 **B Broader Impacts and Limitations**

795 As machine learning continues to advance, tackling the challenges of OOD detection has become
796 essential for ensuring robust and reliable model performance in real-world applications. We introduce
797 a median-centric framework for OOD detection that enhances OOD handling and model safety.
798 The impact of our research goes beyond theoretical advancements, with practical applications in
799 healthcare, autonomous systems, and finance. By improving OOD detection, we address a key
800 challenge in model deployment, fostering greater trust and adoption of machine learning technologies.
801 Future work could explore integrating it with generative models for outlier synthesis or adapting it to
802 dynamic environments where OOD distributions evolve over time.

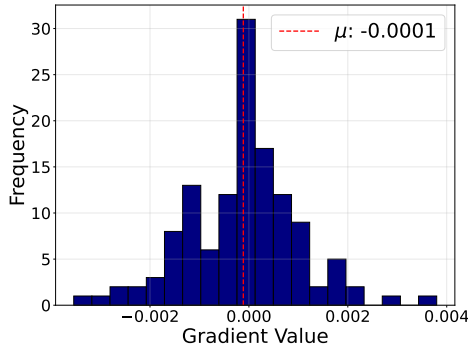
803 A potential limitation of Medix lies in its reliance on a mixture of unlabeled InD and OOD data,
804 which, in practice, may be noisy, corrupted, or inconsistently sampled. Three pieces of evidence in
805 this paper, however, bound the practical impact of this concern. First, the assumption is standard—
806 it mirrors real deployment conditions and is a common starting point in robust learning theory
807 rather than a quirk of our method. Second, our guarantees are explicit and relaxable: the main
808 theorems formalize the required conditions and quantify robustness up to $\pi = 0.5$ contamination,
809 Remark 4.3 states the precise separation condition, and Theorem C.3 extends the guarantee to the
810 weaker finite-fourth-moment setting without sub-Gaussian tails. Third, the empirical resilience is
811 measured directly: removing 15.98% of low-confidence pseudo-labels leaves FPR95 unchanged at
812 0.01% (Appendix A.5), showing the filter tolerates substantial label noise. Together these make the
813 assumptions both interpretable and, where possible, verified in practice.

814 C Proof of Theorem

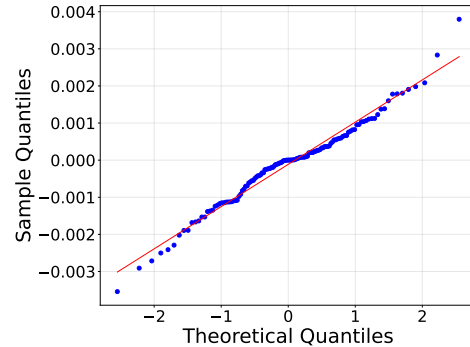
815 To aid clarity and consistency throughout the theoretical and empirical sections, we summarize the
 816 key notation used in this paper in Table 7. This includes definitions for gradient-related quantities,
 817 sample sizes, and filtering terms, as well as parameters that appear in our concentration bounds.
 818 Unless stated otherwise, we use boldface for vectors, $\mathcal{X}$ to denote the input space, and assume all
 819 gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

820 C.1 Bound on the Inlier Misclassification Rate

821 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 822 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 823 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 824 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

825 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
 826 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\nabla}_{\text{in}}$. We define
 827 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
 828 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
 829 than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
 830 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict
 831 optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification
 832 rate, ERR_{in} .

833 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 834 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 835 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

836 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

837 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

838 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

839 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

840 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 841 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

842 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

843 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 844 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

845 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

846 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 847 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

848 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

849 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

850 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
851 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

852 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

853 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

854 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

855 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

856 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
857 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
858 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
859 called *good inliers*.

860 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 861 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

862 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 863 thereby contradicting the optimality of $\mathcal{S}^*$.

864 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 865 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

866 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

867 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

868 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

869 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 870 available for swapping in.

871 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 872 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

873 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

874 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

875 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

876 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

877 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 878 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

879 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

880 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

881 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

882 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

883 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

884 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
885 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

886 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
887 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
888 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

889 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

890 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
891 increase it, so the median distance from μ_j does not increase and possibly decreases.

892 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

893 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

894 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
895 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

896 □

897 C.2 Bound on the Outlier Retention Rate

898 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
899 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
900 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
901 between the OOD and InD gradient means.

902 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 903 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

904 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

905 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

906 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

907 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

908 *Proof. Step 1: Concentration of OOD Gradients.*

909 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 910 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

911 has probability at most $p_{\text{out}}(\varepsilon)$.

912 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

913 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

914 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

915 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

916 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

917 which holds with probability at least $1 - \delta$.

918 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

919 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

940 **Differences with the original bound.** The original bound, which assumes sub-Gaussianity, features
 941 a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

942 that decreases exponentially with dimension d , making the bound extremely tight for large d . In
 943 contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay
 944 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

945 that decays as $O(1/d)$, yielding a looser bound for larger d . The original bound requires sub-Gaussian
 946 tails, while the new one only assumes finite fourth moments, allowing for heavier tails. In particular,
 947 both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

948 that shrinks as m_{in} increases, independent of the tail assumptions.

949 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 950 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 951 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 952 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 953 bound derived from the finite fourth moment assumption.

954 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

955 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 956 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

957 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 958 S_{wild} .

959 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 960 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

961 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

962 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

963 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

964 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

965 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

966 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

967 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

968 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
969 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

970 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
971 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

972 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

973 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

974 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

975 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 976 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

977 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 978 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

979 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 980 thereby contradicting the optimality of $\mathcal{S}^*$.

981 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 982 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

983 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

984 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

985 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

986 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 987 available for swapping in.

988 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 989 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

990 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

991 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

992 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

993 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

994 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 995 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

996 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

997 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

998 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

999 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

1000 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

1001 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 1002 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

1003 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 1004 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 1005 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

1006 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

1007 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 1008 increase it, so the median distance from μ_j does not increase and possibly decreases.

1009 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

1010 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

1011 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 1012 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

1013 □